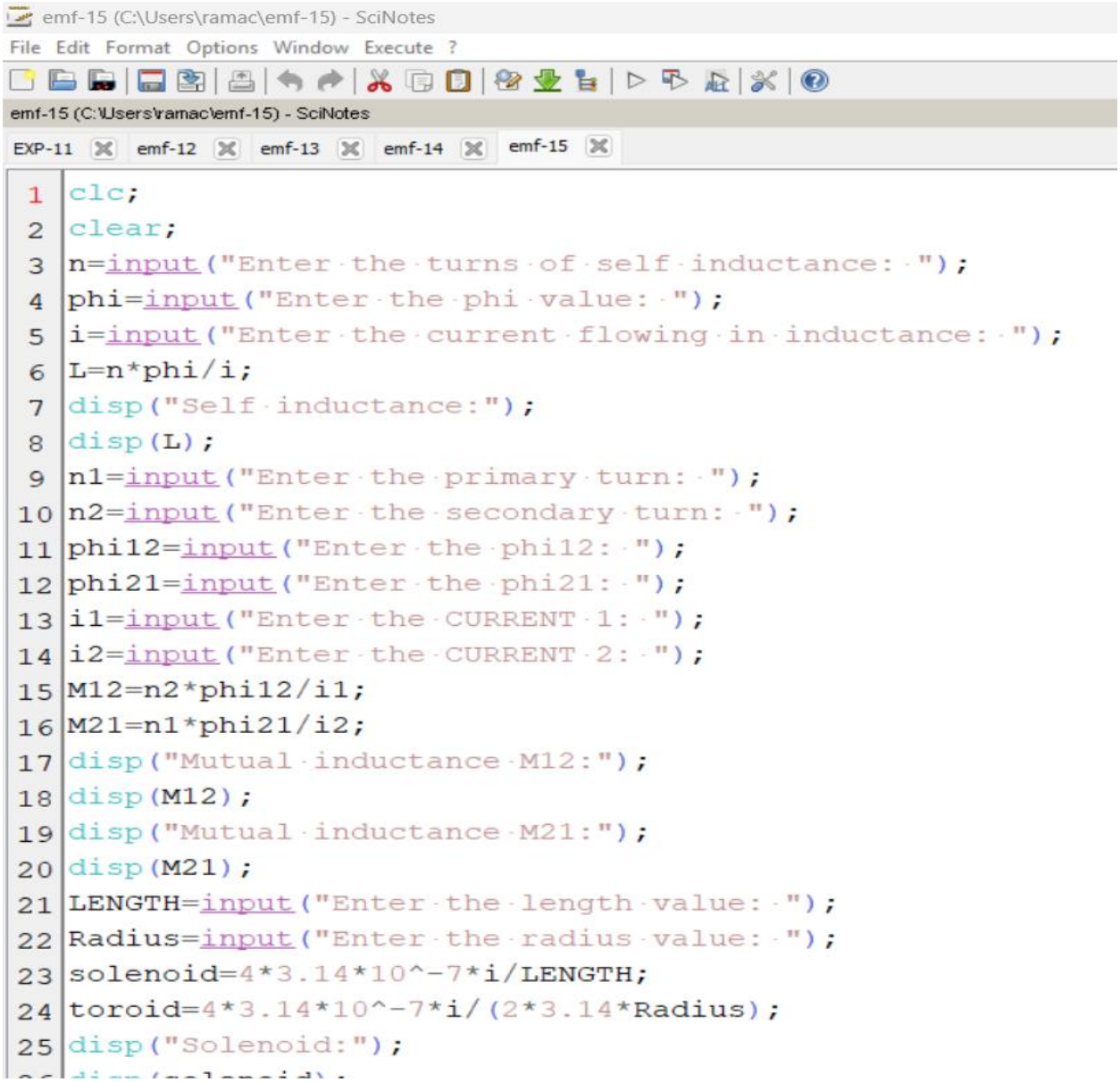
Exp.no-15 Determination and plotting of Inductance (i) Self-inductance and Mutual inductance of current carrying coil, (ii) solenoid and toroid with different (a) length and (b) circumferential area.

Aim: To determine and plot Inductance (i) Self-inductance and Mutual inductance of current carrying coil, (ii) solenoid and toroid with different (a) length and (b) circumferential area.

Software required: SCILAB version 6.0.1

Practical output:



Scilab 6.1.1 Console

File Edit Control Applications ?

Scilab 6.1.1 Console

```
Enter the phi12: 4

Enter the phi21: 5

Enter the CURRENT 1: 3

Enter the CURRENT 2: 5

   "Mutual inductance M12:"

    333.33333

   "Mutual inductance M21:"

    200.
Enter the length value: 5

Enter the radius value: 2

   "Solenoid:"

    0.0000008

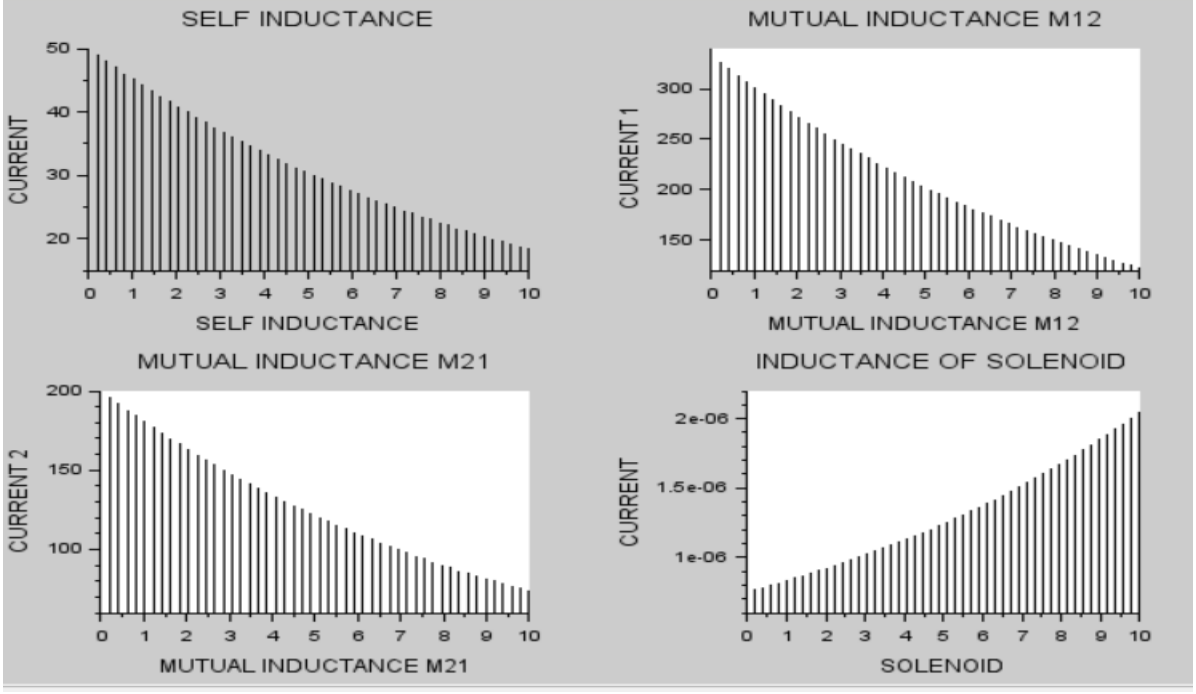
   "Toroid:"

    0.0000003
```

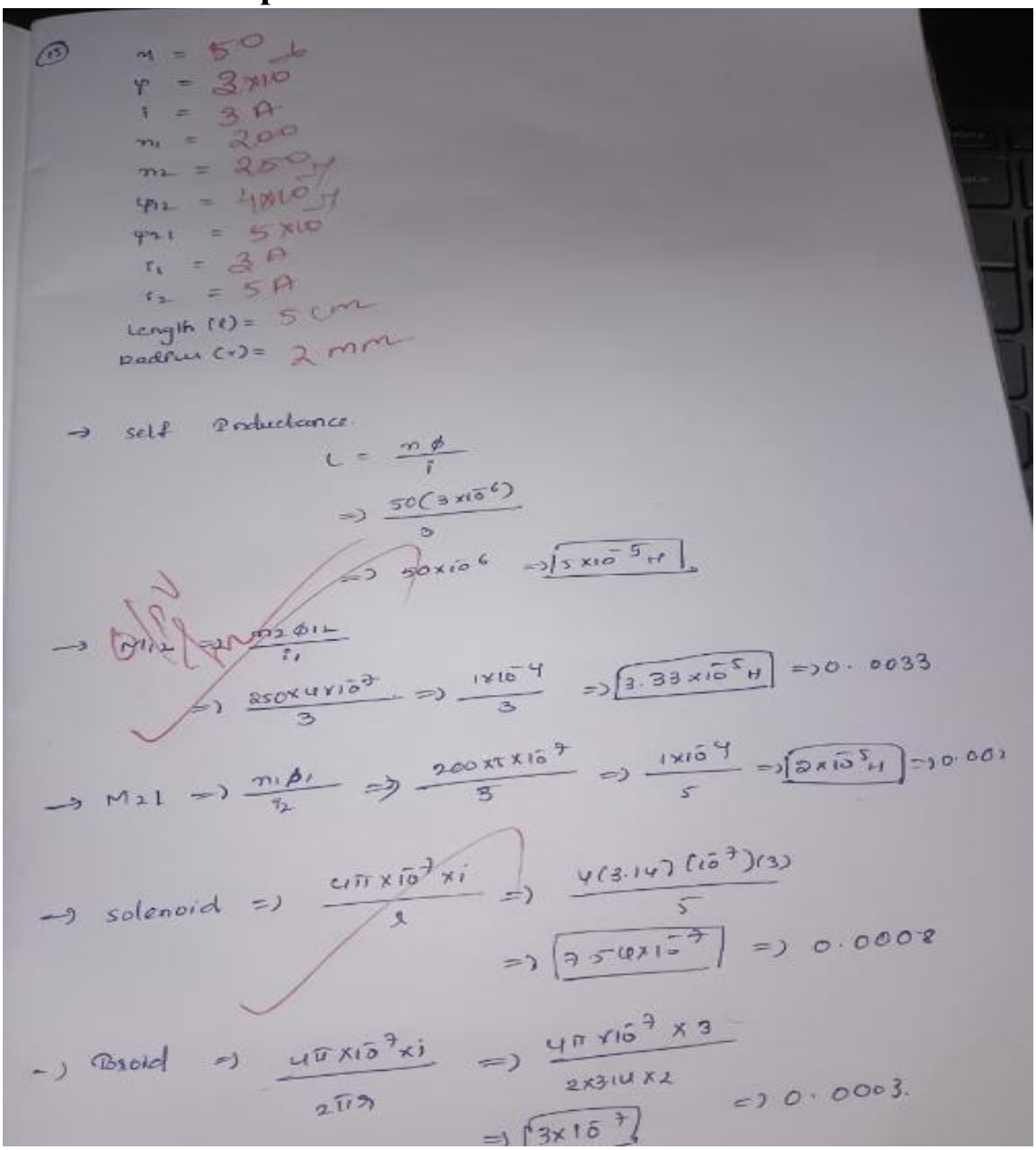
Graphic window number 0

File Tools Edit ?

Graphic window number 0



Theoritical Output:



Result :

Thus determination and plotting of Inductance ,Self-inductance, Mutual inductance of current carrying coil, solenoid and toroid with different (a) length and (b) circumferential area Are performed and verified successfully.